Cloud Identification and Properties Retrieval of the Fengyun-4A Satellite Using a ResUnet Model

Zhijun Zhao, Feng Zhang, Qiong Wu, Zhengqiang Li, Xuan Tong, Jingwei Li, and Wei Han

Abstract— **The Advanced Geostationary Radiation Imager (AGRI) onboard the Fengyun-4A (FY4A) satellite has good cloud observation ability, but it still absents all-weather and high-precision official cloud products. This study develops a deep-learning ResUnet model for all-weather retrieval of cloud phase (CLP) and cloud properties using the brightness temperature from water vapor and longwave infrared channels of AGRI. The ResUnet model is trained with the Himawari-8 satellite Level-2 (H8-L2) cloud products as true targets and adopts image-by-image way to learn the spatial structure information of clouds, which compensates for the difficulty of retrieving thick clouds by thermal infrared radiation at night to some extent. On an independent testing dataset, the model has an overall accuracy of 90.64% for CLP identification and performs well at retrieving cloud top height (CTH). Even without using visible and near-infrared radiation, the root-mean-square error of cloud effective radius (CER) and cloud optical thickness (COT) estimations still reaches 7.14 μm and 9.01 in the range of 0–60. To further illustrate the reliability and applicability, CLP and cloud properties provided by the Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO) and Moderate Resolution Imaging Spectroradiometer (MODIS) are used as benchmarks to assess the quality of cloud products from FY4A satellite Level-2 (FY4A-L2), H8-L2, and ResUnet model retrieval. The ResUnet model provides a significant improvement over FY4A-L2 for the accuracy of cloud identification and in the quality of CTH products. In the range of 0–40 μm (0–60), the CER (COT) product of ResUnet model retrieval has a reliable and higher precision that is comparable with H8-L2.**

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Index Terms— **Cloud phase (CLP), cloud properties, Fengyun-4A (FY4A) satellite, ResUnet model.**

I. Introduction

CLOUDS are important meteorological elements that play a crucial role in the Earth-atmosphere energy balance and atmospheric water cycle [1], [2], [3], [4]. The influence of clouds on the Earth-atmosphere system and water cycle is not only determined by their macro-physical characteristics but also by their microphysical characteristics [5], [6], such as cloud shape, horizontal extension, life history, cloud top height (CTH), cloud base height, cloud phase (CLP), cloud effective radius (CER), and cloud optical thickness (COT) [7], [8]. Therefore, understanding the macro- and microphysical parameters of large-scale cloud systems is of great significance for systematically tracking how cloud systems evolve under different dynamic conditions, understanding atmospheric physical processes, and improving weather simulation and data assimilation [9], [10], [11].

Satellite remote sensing technology (including that on board geostationary and polar orbit satellites) uses visible light, infrared, and microwave detection instruments to detect and process various scenes on the ground through photography or scanning, information sensing, transmission, and processing. It is widely used in cloud observation and retrieval because of the wide observation range and its lack of space–time constraints [12]. The Himawari-8 (H8) and Fengyun-4A (FY4A) geostationary satellites, the CloudSat/Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO), and the Aqua polar orbiting satellites are notable examples that provide series of cloud products.

At present, common cloud detection and CLP identification methods are dominated by the traditional threshold methods and machine learning (ML) algorithms [13], [14], [15], [16]. Based on the threshold of reflection spectra differences between clouds and underlying surfaces using visible and shortwave infrared channels, the Landsat 8 Operational Land Imager and the Moderate Resolution Imaging Spectroradiometer (MODIS) achieved efficient automatic cloud detection with an accuracy of over 85% [17]. CLP can be accurately identified by setting the reflectance difference threshold for two shortwave infrared channels (central wavelength at 1.6 and 2.1 μm) and the brightness temperature (BT) difference threshold for two longwave infrared channels (central wavelength at 8.5 and 11 μm, or 11 and 12 μm) [18], [19], [20]. However, there

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is no unified standard in the traditional threshold methods, and many assumptions and prior knowledge are needed to determine the threshold, making the accuracy is apt to reach the upper limit [21]. DeepMask, a novel cloud detection algorithm for optical satellite remote sensing image based on the residual neural network (Resnet), improved the average accuracy of cloud detection to 93.56% on the Landsat 8 cloud cover validation dataset with eight different terrain types [22]. In addition, CLP can be identified with an accuracy of over 91% using multichannel data from the visible/infrared radiometer of the Fengyun-3A (FY3A) satellite via a supervised back-propagation neural network and an unsupervised self-organizing feature map neural network, indicating that the neural network approach is superior to the threshold technology of the current FY3A running products [23]. Therefore, the ML algorithms can automatically obtain the contour features of the clouds based on massive satellite data, make up for the defects of threshold methods to a certain extent, and greatly improve the accuracy and efficiency of cloud detection and CLP identification.

Due to the limitation of observation time in different satellite spectral channels (visible, near-infrared, and thermal infrared), the Comprehensive Analysis Program for Cloud Optical Measurement for H8/Advanced Himawari Imager (AHI) only effectively retrieves the COT and CER for ice and water clouds by combining the lookup table method during the daytime [24], [25]. To obtain the cloud properties in nighttime, Wang et al. [26] input the MODIS spectral observations from multiple infrared channels from 6.7 to 15.0 μm into the efficient radiative transfer model and simultaneously retrieved the ice COT, CER, and CTH during the daytime and nighttime. Consistent daytime and nighttime retrievals enable diurnal cycle samples of global ice clouds, but only thin ice clouds ($0.5 < \text{COT} < 7$ and $\text{CER} < 50$ μm) have the better retrieval results [26]. Moreover, traditional infrared split-window method and infrared-based optimal estimation are limited in the retrieval of optical thick cloud ($\text{COT} > 10$) due to the blackbody principle [27], [28]. To further make up for the shortcomings of traditional algorithms for cloud properties, Minnis et al. [29] used a neural network method to estimate nocturnal opaque ice COT using radiances from MODIS multispectral infrared channels (central wavelength at 3.7, 6.7, 11.0, and 12.0 μm). Training and validation results gave an ice COT estimate up to 150, and the correlation coefficients was higher than 69% [29]. Yang et al. [30] also used the H8/AHI infrared data (central wavelength at 3.9–13.3 μm) to obtain the unified cloud top temperature (CTT) and CTH retrieval at high spatial (0.02°) and temporal (10 min) resolution during day and night over East Asia via a XGBoost algorithm. Comparing with the official cloud product algorithms of MODIS and Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP), the XGBoost algorithm had a good performance for CTH and CTT retrieval. Moreover, Wang et al. [31] used a deep neural network (DNN) to achieve all-weather CTH and ice cloud COT retrieval in East Asia from H8/AHI infrared BT. Comparing it with the results from the two physics retrieval models, DNN model extended the predictive ice COT to 200, and the relative deviation is about 20% for high ice cloud with $\text{COT} > 1$. Wang et al. [32] developed a convolutional neural network that can consistently retrieve all-weather cloud properties from passive satellite observations without auxiliary atmospheric parameters, and has stability in the diurnal cycle.

Through the above analysis, we find that ML algorithms can use the thermal infrared radiometry to perform cloud retrieval during the daytime and nighttime. However, the retrieval results are better only for thin clouds and more uncertain for thick clouds due to their less sensitive to radiation at night. Moreover, most of the current ML algorithms take polar-orbit satellite cloud products as true target for learning [33]. The polar-orbiting satellites were unable to obtain continuous diurnal variations of clouds over the entire observation area due to their fixed transit time, so most ML algorithms adopt the way of pixel-by-pixel and image-by-pixel for learning, which results in heavy loss of cloud space structure information. To make the ML model can obtain more spatial information and connection of clouds as much as possible, we adopt the image-by-image way for learning. In addition, the convolution operation of the model can better associate the surrounding spatial information from horizontally inhomogeneous clouds with the thick cloud at night, which to some extent makes up for the defect that the thermal infrared radiometry cannot accurately retrieve thick clouds at night.

As a new generation of geostationary satellites, both H8/AHI and FY4A/Advanced Geostationary Radiation Imager (AGRI) have carried out continuous and large-scale cloud observation in East Asia (especially in China), among which H8 satellite only provides cloud products during the daytime. Although FY4A satellite provides abundant cloud products (such as cloud detection, CLP, CTT, and CTH) during the daytime and nighttime, the official is unable to provide cloud optical and microphysical property products (such as COT and CER), and the quality of relevant cloud products (such as CLP and CTH) needs to be tested further. Therefore, this study develops a deep-learning ResUnet model to realize image-by-image and all-weather retrieval for FY4A satellite CLP, COT, CER, and CTH products using AGRI water vapor and longwave infrared BT data as input features, and daytime CLP, CTH, CER, and COT products from H8 satellite Level-2 (H8-L2) products as true targets. Finally, the quality and reliability of cloud products from FY4A satellite Level-2 (FY4A-L2), H8-L2, and the ResUnet model retrieval are evaluated and compared using CLP and cloud properties products provided by CALIPSO and MODIS as benchmarks. This study is organized as follows. Section II describes the data sources. Section III introduces the analytical methods and the ResUnet model. Section IV presents the results and discussion, and Section V presents the conclusions.

II. Data Sources

A. Satellite Data

The FY4A satellite was successfully launched by China in December 2016, and the onboard AGRI instrument was the first to achieve the full-disk imaging of the Earth using autonomous 2-D scanning technology. Full-disk

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scanning takes 15 min to perform each time, and AGRI has 14 channels that include visible and near-infrared, shortwave infrared, mediumwave infrared, water vapor, and longwave infrared bands, with spectral wavelength ranges from 0.45 to 13.8 μm (http://data.nsmc.org.cn/PortalSite/Data/Satellite.aspx) [34]. In this study, we used BT data from water vapor and longwave infrared channels measured at 6.25, 7.1, 8.5, 10.8, 12.0, and 13.5 μm (bands 9–14) with a spatial resolution of 4 km in 2019 and 2020 as input features of the ResUnet model. In addition, we evaluated and discussed the quality and precision of CLP and CTH products from FY4A-L2.

H8 is a new-generation geostationary meteorological satellite launched by Japan in 2014. It is equipped with an AHI with 16 visible and infrared channels covering 0.47–13.3 μm, providing observational data with different spatial (0.5–2 km) and temporal (2.5–10 min) resolutions [35]. We used quality assurance (QA) and three cloud properties (COT, CER, and CTH) data from H8-L2 in 2019–2020 as true targets of the ResUnet model. It should be noted that QA can be converted to CLP. The quality and precision of these cloud products, which have a spatial resolution of 5 km and a temporal resolution of 10 min, were also be evaluated and discussed. More detailed information about H8-L2 cloud products can be obtained at https://www.eorc.jaxa.jp/ptree/index.html.

The Aqua (launched in May 2002) and CALIPSO (launched in April 2006) satellites are two polar-orbiting satellites in the A-Train series of satellites launched autonomously by the United States [36]. The detection spectrum of the MODIS carried by Aqua is distributed across 36 spectral bands ranging from 0.4 to 14.4 μm, which can obtain radiation information about clouds, aerosols, atmosphere, and surface (https://ladsweb.modaps.eosdis.nasa.gov/) [37]. The CALIOP onboard CALIPSO is a dual-wavelength (532 and 1064 nm) polarization-sensitive spaceborne lidar that can be used to detect high vertical resolution profiles of clouds and aerosols (https://subset.larc.nasa.gov/calipso/login.php) [38]. In this study, we used datasets from two sources in January, April, July, and October 2020 as benchmarks when evaluating and comparing the data quality and precision of cloud products from H8-L2, FY4A-L2, and the ResUnet model retrieval after spatial and temporal matching in the whole year and different seasons: CLP and CTH information obtained from the feature classification flags and layer top altitude profile of CALIPSO Level-2 cloud layer product (CAL_LID_L2) with a horizontal resolution of 1 km, and cloud properties (CTH, CER, and COT) products obtained from MODIS Level-2 version 6.0 (MYD06) products with a horizontal resolution of 1 km. It should be noted that the CAL_LID_L2 product does not provide CER and COT information.

A description of the satellite data used in this study is given in Table I. There are some differences in the spatial and temporal resolutions of different satellite sensors, so data preprocessing is necessary. In addition, considering the influence of the satellite zenith angle (SZA) and surface conditions on the radiation budget and energy balance, we added the following into the ResUnet model: SZA of FY4A/AGRI and eight-day average surface emissivity (SE) information from the Terra satellite (launched in December 1999) MODIS Level-1B 6.1 version product (MOD11C2). The MOD11C2 product is obtained from bands 20, 22, 23, 29, 31, and 32 at 3.75, 3.96, 4.05, 8.55, 11.03, and 12.02 μm, respectively, with a spatial resolution of 5 km.

B. ERA5 Reanalysis Data

The ERA5 reanalysis data are a comprehensive global atmosphere, land, and ocean dataset obtained by European Centre for Medium-Range Weather Forecasts (ECMWF) through predictive models and data assimilation systems, which assimilates as much observation data as possible of the upper atmosphere and near surface from 1979 to the present (https://cds.climate.copernicus.eu) [39]. Considering the influence of meteorological fields like temperature and humidity on cloud formation and evolution, we added the following data into the ResUnet model: hourly surface temperature (ST) at 2 m, total column water vapor (TCWV), air temperature profile (ATP), and relative humidity profile (RHP) with a spatial resolution of 0.25°. Water vapor condenses and rises to form clouds. In this process, TCWV and RHP provide sufficient water vapor and humidity conditions for cloud formation, and ST and ATP provide sufficient dynamic conditions for cloud vertical rise and horizontal movement. These meteorological conditions are crucial to the cloud height and optical properties, so the four meteorological fields should be taken as input features in the cloud retrieval model. Here, we selected only the profile data at four pressure levels of 300, 500, 850, and 1000 hPa.

C. Study Area

The observation region of FY4A/AGRI was 24.12°E–185.28°E and 80.57°S–80.57°N, and each image was composed of 2748 × 2748 nonuniform grids with a spatial resolution of 4 km at nadir. However, the observation region of H8/AHI was 80°E–200°E and 60°S–60°N, and each image was composed of 2401 × 2401 uniform grids with a spatial resolution of 0.05°. Taking into account the differences between the observation areas of the two satellite sensors and the production of the training and testing datasets in the ResUnet model, we ultimately selected East Asia with longitude ranging from 80°E to 182.4°E and latitude ranging from 42.4°S to 60°N as the study area, which includes 2048 × 2048 uniform grids with a spatial resolution of 0.05° (Fig. S1).

III. Methodology

A. Data Preprocessing

During model training and testing, it is necessary to ensure time and space correspondence between the input features and the true targets. However, the data used in this study, including meteorological fields from ERA5 reanalysis, SE from MODIS, nonuniform grid BT from FY4A/AGRI, and uniform grid cloud products from H8-L2, all have a spatiotemporal mismatch and nonnormal distribution. Therefore, these grid data

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TABLE I
DESCRIPTION OF THE USED SATELLITE DATA

Satellite Name	Satellite Sensor	Wavelength Range	Dataset Name	Spatial Resolution	Temporal Resolution
FY4A	AGRI	0.45–13.8 μm	L1: GEO_SDR L2: CLP/CTH/CLM	4 km	15 min
H8	AHI	0.47–13.3 μm	L2: CLP/CTH/CER/COT	5 km	10 min
CALIPSO	CALIOP	532&1064 nm	CAL_LID_L2: CLP/CTH	1 km	---
Aqua	MODIS	0.40–14.4 μm	MYD06: CTH/CER/COT	1 km	---
Terra			MOD11C2: SE	5 km	8 days

need to be preprocessed before being input into the ResUnet model.

Spatial interpolation methods are widely used in data processing, such as the inverse distance weighting (IDW), Kriging, and radial basis function (RBF) neural network interpolation methods. The Kriging interpolation method requires that the data conform to a normal distribution, and the IDW method only considers the distance between the known sample point and the point to be interpolated without considering the influence of other factors, resulting in limitations in practical applications [40], [41]. However, the RBF neural network interpolation method can make up for these shortcomings and determine the corresponding network topology based on the specific problem. It also has a fast convergence speed and good generalization ability in dealing with nonlinear problems [42]. Python and MATLAB provide corresponding packages for implementing these three interpolation methods.

Based on the hourly H8-L2 cloud products data format, we used the RBF neural network interpolation method to interpolate meteorological fields from ERA5 reanalysis and BT from FY4A/AGRI data into a uniform grid with a spatial resolution of 5 km at the corresponding moment. The SE data from MODIS were then interpolated into a uniform grid and into the same channels as FY4A/AGRI. The model dataset was being further constructed after data preprocessing, as explained further in Section III-C.

B. ResUnet Model

The deep-learning ResUnet model used in this article is a combination of a semantic segmentation network and Resnet. Semantic segmentation is an important branch in the field of image processing and machine vision, in which Unet is the most common and simplest segmentation model [43]. The Unet model is beneficial in biomedical image segmentation and has been widely used in the field of remote sensing in recent years [44], [45]. Before Resnet was proposed, all neural networks were superimposed by a convolution layer and a pooling layer. It was believed that having more layers of convolution and pooling allowed for more image feature information to be obtained. However, it was found that the accuracy of the training dataset would drop significantly with the deepening of the network, and problems like gradient disappearance, gradient explosion, and gradient degradation would also occur in actual experiments [46], [47]. Therefore, He et al. [48] proposed that these problems could be solved by using a batch normalization (BN) layer and weakening the strong connections between layers in the Resnet structure.

The conceptual diagram of the ResUnet model architecture constructed in this study is shown in Fig. 1. The model is U-shaped and composed mainly of encoding and decoding [Fig. 1(a)]. During encoding portion, we convert an input image with x channels and a size of 64×64 into an intermediate vector with 2048 channels and a size of 8×8 through convolution and pooling. STAGE1–STAGE4 consists of two bottlenecks (BTNK1 and BTNK2) with similar structures. These two bottlenecks illustrate the nature of the residual network, where C, H, W, and S are the channel number, height, width, and step, respectively [Fig. 1(b)]. The internal structure of the decoding portion is shown in Fig. 1(c). From STAGE5–STAGE8, we can obtain a vector with a shape of $64 \times 64 \times 64$ when the intermediate vector is upsampled and convolved. Finally, we can obtain the output image with n channels and a size of 64×64 after processing with a 1×1 convolution kernel.

It is worth noting that each STAGE in the decoding portion is connected to a corresponding STAGE in the encoding portion through a concatenate layer. This kind of skip connection can allow the network to retain more original image features. It should also be noted that CLP retrieval is essentially a multiclassification problem, and the input and output images have 23 features and four channels, respectively, while the retrieval of COT, CER, and CTH is essentially a regression problem. We also add CLP information into the training and testing processes, resulting in 24 features for input and one channel for output images ($x = 24$ and $n = 1$).

C. Training, Testing, and Assessment of the ResUnet Model

Fig. 2 shows the research framework and flowchart for this study, which is divided primarily into two parts. During data preprocessing, 23 features and four targets are obtained after

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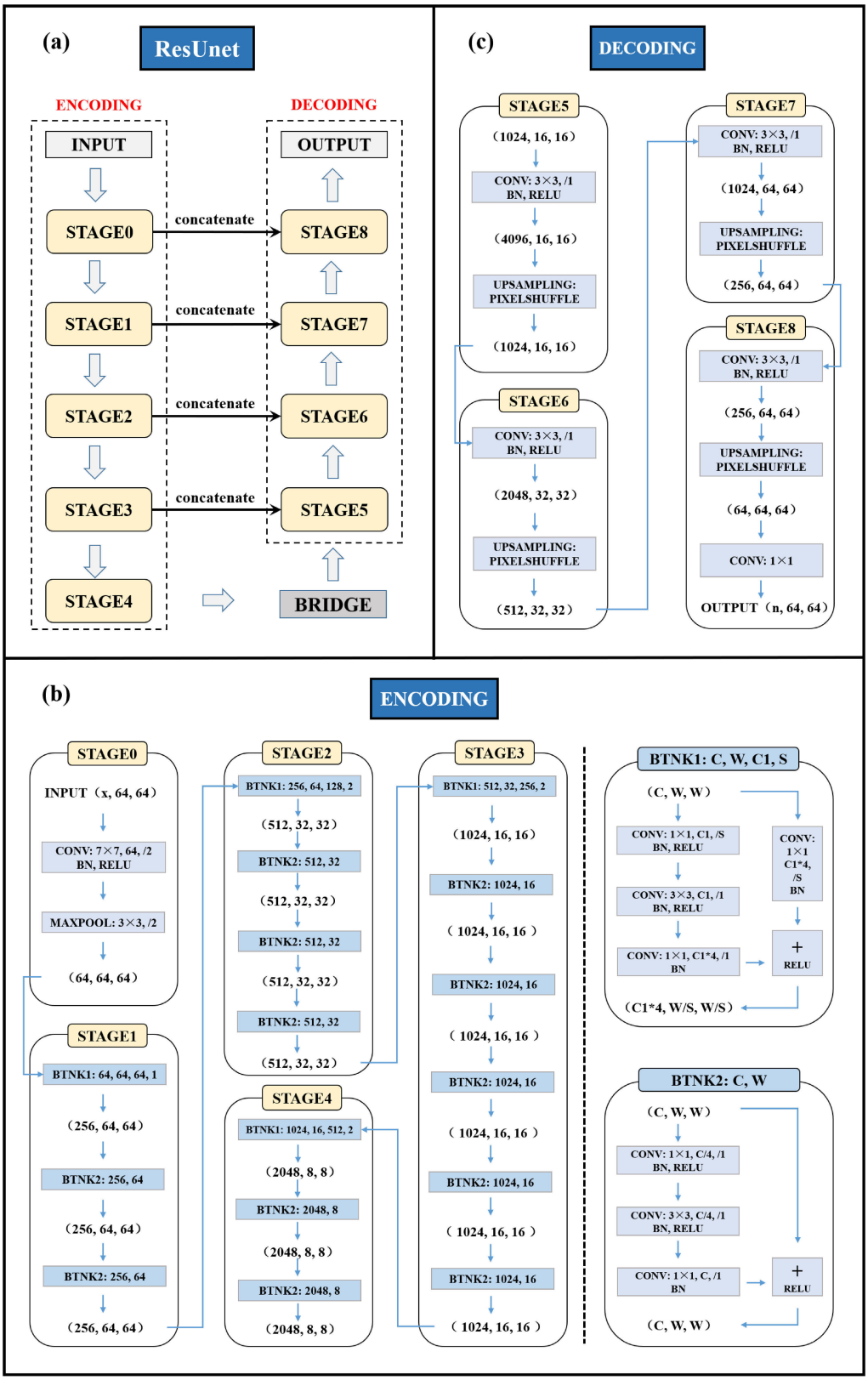


Fig. 1. Conceptual diagram of the (a) ResUnet model architecture; (b) encoding structure; and (c) decoding structure, where C, H, W, and S are the number of channels, height, width, and stride, respectively (ReLU is the activation function, BN represents the batch normalization, and /1 and /2 represent the stride is 1 and 2, respectively).

the spatiotemporal matching of various data. Since the full image size of 2048 × 2048 is too large, placing it directly into the model will overload the memory and reduce computational efficiency. Before applying the dataset to the model, we cut the full image into 1024 smaller pieces with sizes of 64 × 64 to use as samples. Moreover, we need to further screen out

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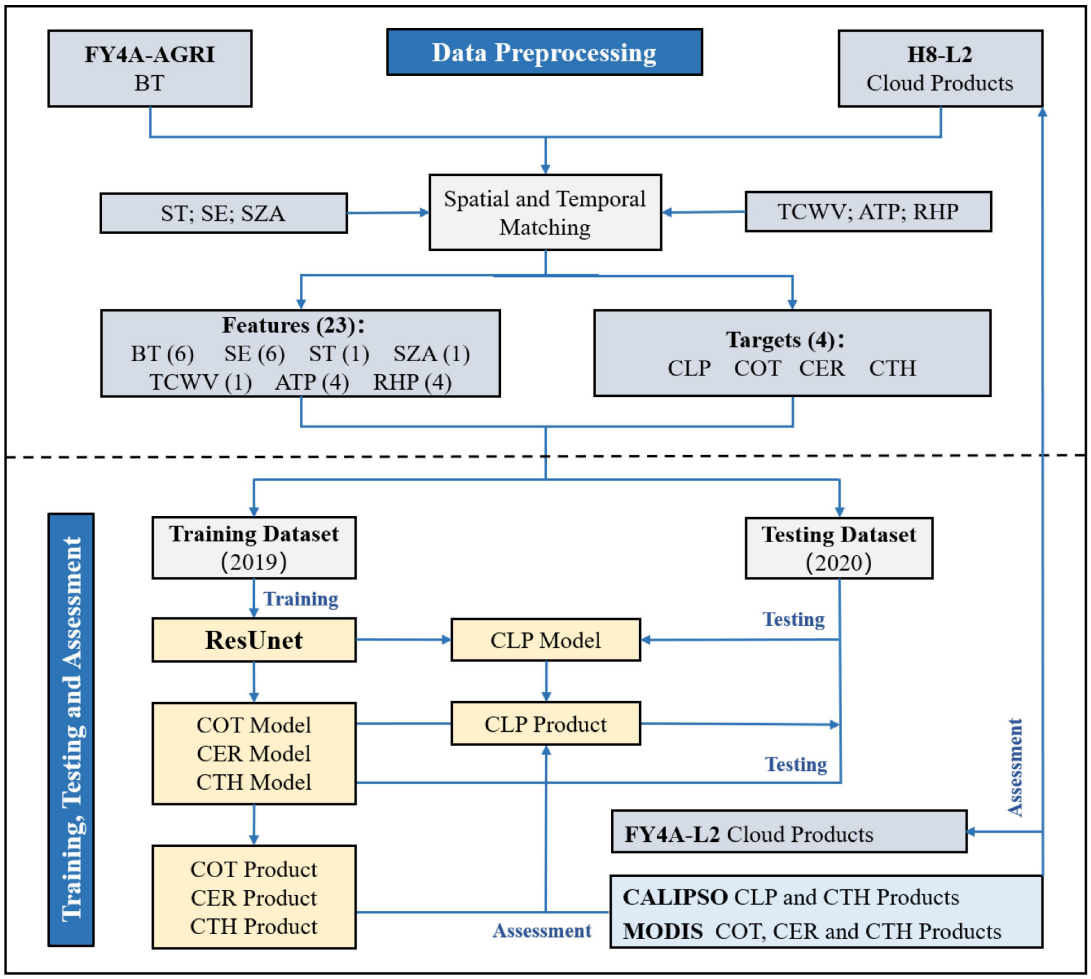


Fig. 2. Research framework and flowchart of this study.

the samples in which missing values and low confidence pixels constitute more than 10% of the full image. Finally, we obtain 374 587 and 342 566 valid samples from the training dataset in 2019 and the testing dataset in 2020, respectively. According to the spatial distribution of valid samples in Figure S1, land produces more valid samples than oceans, while there are almost none when SZA is larger than 70°.

The sample number and parameter settings of the four cloud models are shown in Table II. To improve the learning ability of the CLP model for mixed or uncertain clouds as much as possible, 159 223 and 50 000 samples with the occurrence of mixed or uncertain cloud pixels were selected from the valid samples for training and testing in the CLP model. However, in the models for CTH, COT, and CER, the number of samples for training and testing was significantly reduced and varied due to the removal of samples containing clear sky pixels. All testing samples were randomly selected to ensure that they were representative. In the parameter settings of the model, we uniformly used 400 epochs and a learning rate of 0.001. The CLP model has a larger number of samples and involves a multiclassification problem, so the batch size and loss function used were set to 128 and CrossEntropyLoss, while these were 64 and MSELoss for the other three models.

In the training, testing, and assessment portions, we first obtain the CLP product by using the ResUnet model. Then, we add it into the CTH, COT, and CER models for training and testing because of the effective improvement that CLP information provides for the retrieval of cloud properties (Table III). Finally, we assess the quality of cloud products from FY4A-L2, H8-L2, and ResUnet model retrieval using the cloud products from CALIPSO satellite and MODIS in January, April, July, and October 2020 as benchmarks.

For the CLP products, we evaluate the precision and ability of the model in identifying different CLP by calculating the overall accuracy (OA) using (1) and constructing a confusion matrix. In addition, to evaluate the CLP model's effect on cloud detection, we also need to calculate evaluation indicators, such as precision, recall, and F1 score using (2)–(4). The F1 score is the weighted harmonic average of precision and recall, and is a measure of the precision of the binary classification model, where the best value is 1 and the worst value is 0. For the COT, CER, and CTH products, we mainly plot a joint probability density distribution map and calculate the mean absolute error (MAE), mean bias error (MBE), and root-mean-square error (RMSE) to evaluate the quality of

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TABLE II
SAMPLE NUMBER AND PARAMETER SETTING OF FOUR CLOUD MODELS

Cloud Model	Sample Number		Parameter Setting			
	Training Dataset (2019)	Testing Dataset (2020)	Batch Size	Epoch	Learning Rate	Loss Function
CLP	159223	50000	128	400	0.001	CrossEntropyLoss
CTH	57048	24576	64	400	0.001	MSELoss
CER	36140	24576	64	400	0.001	MSELoss
COT	76455	24576	64	400	0.001	MSELoss

TABLE III
RELATIVE IMPORTANCE EVALUATION OF EACH INPUT VARIABLE

Cloud Model	Evaluation Indicator	**ResUnet**	ResUnet with BT	Ablated Variable								
				BT_{WVC}	BT_{LWC}	SE	ST	TCWV	RHP	ATP	SZA	CLP
CLP (%)	OA	**90.62**	89.53	89.30	86.21	90.53	90.55	90.62	90.52	90.68	90.04	---
CTH (km)	RMSE	**0.95**	1.17	0.98	1.08	0.96	0.97	0.95	0.99	0.98	0.99	1.00
	MAE	**0.71**	0.93	0.73	0.82	0.72	0.72	0.72	0.74	0.74	0.75	0.75
	MBE	**0.03**	0.21	0.04	-0.05	0.00	0.06	0.02	-0.04	-0.10	-0.03	0.03
CER (μm)	RMSE	**9.84**	10.89	9.85	10.04	9.93	9.86	9.84	10.01	9.99	10.01	10.53
	MAE	**7.15**	7.96	7.15	7.35	7.20	7.16	7.15	7.29	7.29	7.31	7.62
	MBE	**0.06**	-0.17	0.16	-0.02	-0.15	-0.37	-0.03	-0.33	0.07	-0.18	0.08
COT	RMSE	**19.12**	20.25	19.37	19.54	19.23	19.22	19.15	19.32	19.50	19.33	19.45
	MAE	**11.54**	12.52	11.82	12.04	11.65	11.66	11.58	11.70	11.87	11.77	11.90
	MBE	**-1.76**	-2.57	-1.21	-2.17	-1.55	-1.43	-1.92	-1.27	-1.00	-1.08	-1.38

Note that ResUnet with BT is a model using brightness temperature at bands 9, 10, 11, 12, 13, and 14 from AGRI; BT_{WVC} is the brightness temperature at water vapor channel (bands 9 and 10) from AGRI; BT_{LWC} is the brightness temperature at longwave infrared channel (bands 11−14) from AGRI; Black bold font shows the results of the ResUnet model.

model retrieval. These indicators are calculated as follows:

$$\mathrm{OA} = (\mathrm{TP}+\mathrm{TN})/(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}) \quad (1)$$

$$\mathrm{Precision} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}) \quad (2)$$

$$\mathrm{Recall} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) \quad (3)$$

$$\mathrm{F1} = (2\times \mathrm{Precision}\times \mathrm{Recall})/(\mathrm{Precision}+\mathrm{Recall}) \quad (4)$$

where TP is the number of true classes predicted to be positive classes (true positive), FP is the number of false classes predicted to be positive classes (false positive), TN is the number of true classes predicted to be negative classes (true negative), and FN is the number of false classes predicted to be negative classes (false negative).

IV. RESULTS AND DISCUSSION

A. Importance Evaluation of the Input Variables

To explore the specific impact of each factor on cloud formation and evolution, we evaluate the relative importance

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of each input variable to the model using ablation tests. As shown in Table III, when only BT is input into the CLP model, OA can reach 89.53%. After removing the water vapor channel (BT_{WVC}) and longwave infrared channel BT (BT_{LWC}), OA decreases by 1.32% and 4.41%, respectively, indicating that the relative importance of BT_{LWC} is higher than that of BT_{WVC} (columns 4–6). Among the other influencing factors, the addition of SZA has a certain improvement effect on OA (about 0.6%), but the effect of meteorological fields on CLP retrieval is not obvious (columns 5–10), indicating that BT plays the most important role in CLP retrieval. This is because CLP, as the thermodynamic state of cloud droplets, is highly related to the BT of the observed cloud top, and the absorption of radiation by ice and water particles in the clouds is quite different in the atmospheric window region of thermal infrared bands [49], [50].

Compared with the results from the ResUnet model, the retrieval ability of the CTH, CER, and COT models on the testing dataset drops significantly when only BT is input, which indirectly proves that meteorological fields and SZA are important for the retrieval of cloud properties. Although the BT of water vapor absorption bands is not very sensitive to clouds, water vapor weakens cloud reflectivity less above high clouds, resulting in higher (lower) reflectivity for high cirrus (low clouds). This is probably why BTwvc can improve the retrieval performance of the CTH and COT model (columns 4–6) [51]. We then ablate each variable for the meteorological fields at a specific time to explore the change in performance of the cloud properties model. The results show that TCWV has the least importance, while RHP is the most important. Take the MAE of CTH retrieval as an example, the accuracy of the CTH model will drop by 0.01 and 0.03 km when TCWV and RHP are ablated, respectively (columns 9–10). In particular, the addition of SZA can correct the deviation caused by different observation angles and can reduce the MAE of CTH, CER, and COT retrieval by 0.04 km, 0.16 μm, and 0.23, respectively, which has a favorable impact on cloud retrieval (column 12).

To illustrate the role of CLP in the retrieval of cloud properties, we compare how model accuracy changes after removing CLP (column 13). We find that the CTH, CER, and COT models have a significant drop in accuracy after removing CLP, and MAE will increase by 0.04 km, 0.47 μm, and 0.36, respectively. It shows that the addition of CLP enables the models to better capture the contour features of clouds, which is of great significance for the retrieval of their optical and microphysical parameters.

B. Validation and Assessment of the CLP Model

The validation results on the testing dataset represent the learning ability of the model for true targets from H8-L2. Fig. 3 shows the CLP identification results for 50 000 random samples from the 2020 testing dataset, which are divided primarily into clear sky, water clouds, ice clouds, and mixed or uncertain clouds. In general, the ResUnet model retrieves CLP with an OA of 90.26%. Specifically, the retrieval accuracy of the CLP model for clear sky and ice clouds exceeds 93%,

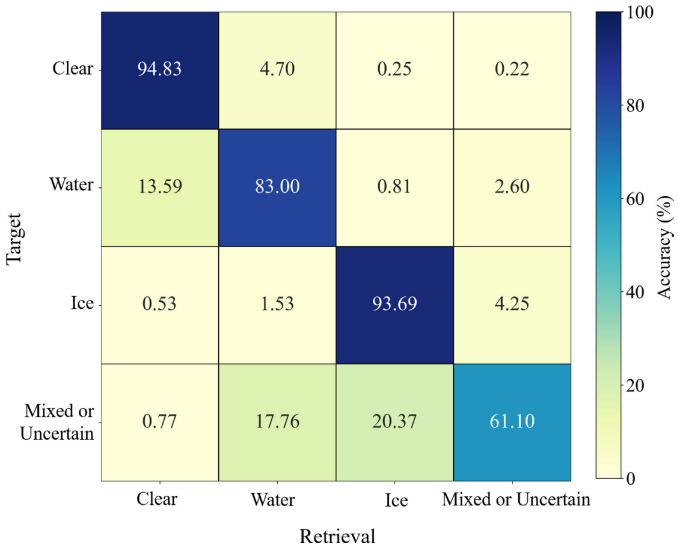


Fig. 3. Confusion matrix of CLP identification for random 50 000 samples from the testing dataset in 2020.

and the retrieval accuracy for water clouds is 83%. However, the CLP model seems unable to fully detect the uncertainty of mixed or uncertain clouds in true targets from H8-L2 data, resulting in an accuracy of 61.10% when identifying mixed or uncertain clouds. Overall, the model performs well in identifying clear sky and ice clouds, clear sky and mixed or unknown clouds, ice clouds, and water clouds.

To better illustrate the applicability and reliability of the CLP model, we take the CLP product from CALIPSO as a benchmark for assessing and comparing the cloud identification performance of that from FY4A-L2, H8-L2, and ResUnet model retrieval. There is the high uncertainty for mixed or uncertain clouds, and mixed clouds are not provided by the official CLP product of CALIPSO. Consequently, we evaluate only the accuracy of recognizing clear sky, water clouds, and ice clouds. Fig. 4 shows the confusion matrix for CLP identification in 2020, where the ordinate and abscissa are the CLP products from CALIPSO and FY4A-L2, H8-L2, or ResUnet model retrieval, respectively. Using the CLP from CALIPSO as a benchmark, the OA of H8-L2 (78.97%) and the ResUnet model (76.74%) is significantly better than that of FY4A-L2 (71.81%). Specifically, the accuracy of FY4A-L2 is only 55.38% in identifying clear sky pixels, and 42.53% of those pixels are misjudged as water clouds, while H8-L2 and the ResUnet model can achieve an accuracy of around 70%. The accuracy of H8-L2 and the ResUnet model when identifying water and ice clouds also slightly outperforms FY4A-L2. Similarly, ResUnet model has a higher OA than FY4A-L2 in different seasons, indicating that the model has a good stability (Table SI).

In addition, we also compare and assess the cloud detection results for the three CLP products by using CALIPSO as a benchmark to conduct a statistical analysis of the four evaluation indicators (Table IV). We find that FY4A-L2 has an OA of 81.77% when recognizing cloudy and clear sky, while H8-L2 and the ResUnet model can reach 86.57% and 84.09%. Precision reflects the proportion of the positive pixels

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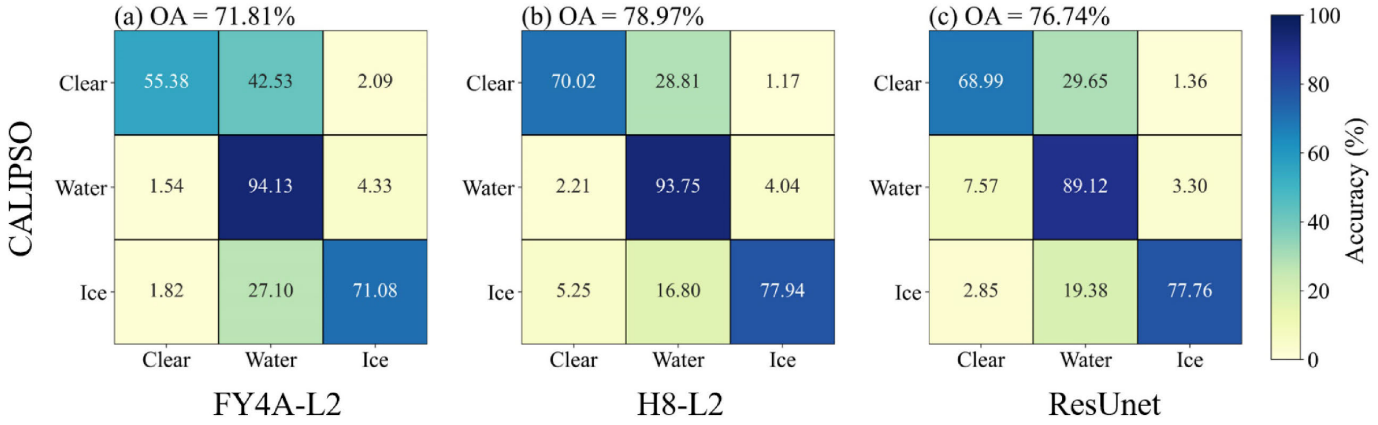


Fig. 4. Confusion matrix of CLP products between CALIPSO and (a) FY4A-L2; (b) H8-L2; and (c) ResUnet model retrieval in 2020.

TABLE IV
EVALUATION OF CLOUD DETECTION FROM CLP PRODUCTS USING CALIPSO DATA IN 2020 AS THE BENCHMARK

Evaluation indicators	FY4A-L2	H8-L2	ResUnet
OA	81.77%	86.57%	84.09%
Precision	96.39%	93.81%	91.51%
Recall	55.38%	70.02%	68.99%
F1	0.70	0.80	0.79

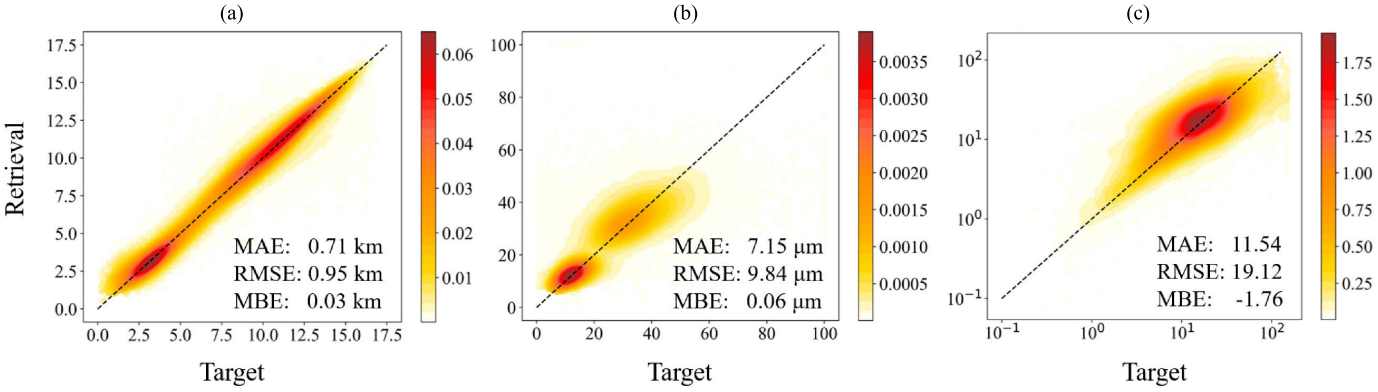


Fig. 5. Joint probability density plots for the (a) CTH; (b) CER; and (c) COT from H8-L2 target and ResUnet model retrieval. Color represents the occurrence frequency obtained by kernel density estimation.

that are determined as true pixels by the model, and recall reflects the proportion of positive pixels that are correctly judged in the total true pixels. We find that the precision of FY4A-L2 (96.39%) is slightly higher than that of H8-L2 (93.81%) and the ResUnet model (91.51%), while the recall of FY4A-L2 (55.38%) is much lower than that of H8-L2 (70.02%) and the ResUnet model (68.99%). Finally, the F1 score gives the cloud detection performance of the three cloud products based on precision and recall. The F1 score for FY4A-L2 is 0.70, which is lower than that of H8-L2 (0.80) and the ResUnet model (0.79). In different seasons, the OA of cloud detection by H8-L2 and the ResUnet model is also higher than that of FY4A-L2 (Table SI). These indicators demonstrate that the performance of cloud detection for the CLP product from FY4A-L2 is improved by the ResUnet model.

The validation and assessment of the three CLP products reveal that FY4A-L2 greatly misjudges clear sky and water clouds. Therefore, the new CLP product dataset from FY4A/AGRI obtained by ResUnet model retrieval can significantly improve the accuracy of cloud detection and CLP identification, and it can provide more accurate CLP products for the next step of CTH, CER, and COT retrieval.

C. Validation and Assessment of the CTH, CER, and COT Models

The validation of the ResUnet model's ability to retrieval cloud properties with the 2020 testing dataset is shown in Fig. 5. The diagonal line represents the location where the retrieval of the ResUnet model is equal to the target. From the joint probability density distribution of CTH in Fig. 5(a),

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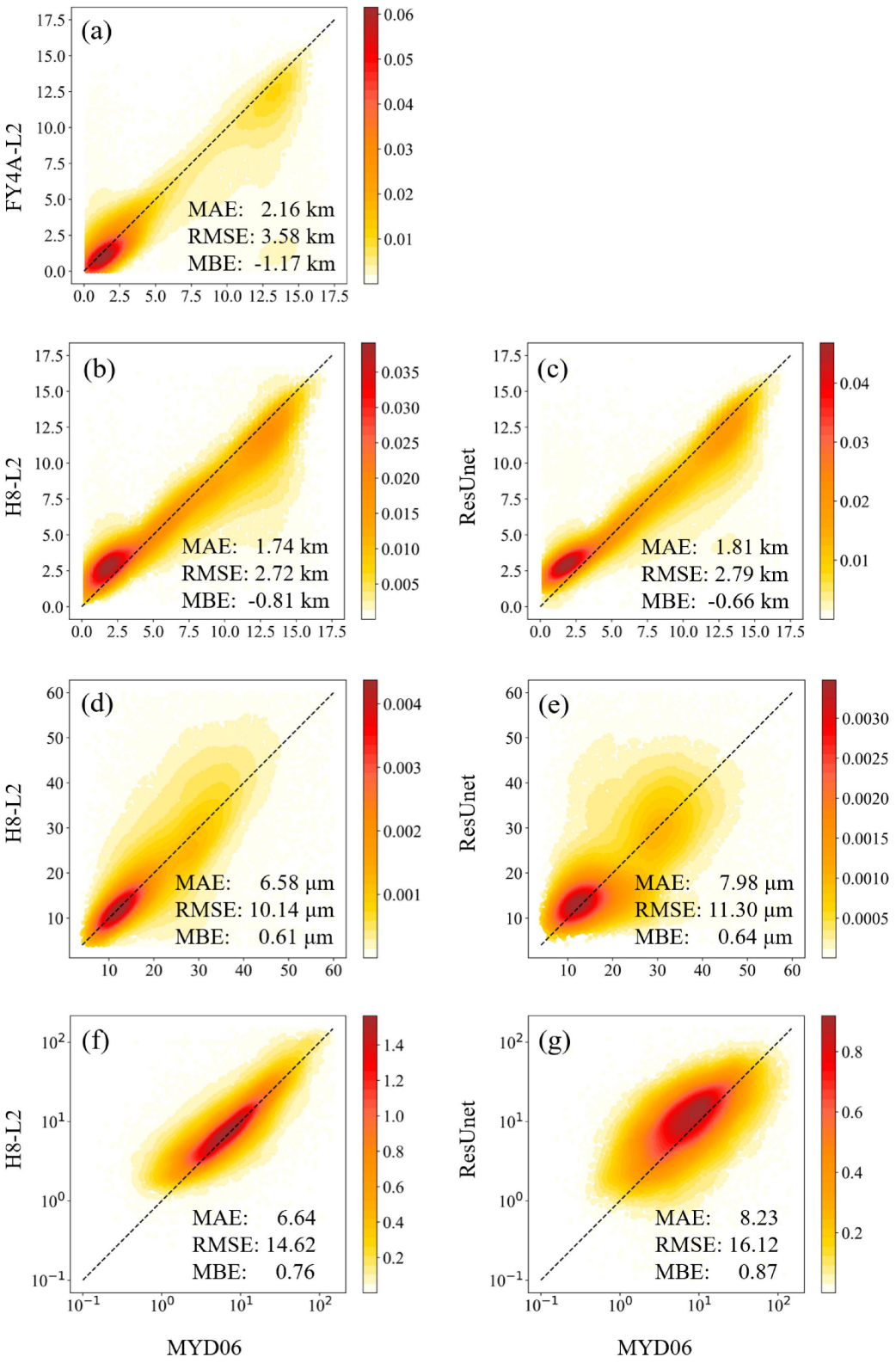


Fig. 6. Joint probability density plots for (a)–(c) CTH; (d) and (e) CER; and (f) and (g) COT products between MODIS and (a) FY4A-L2; (b), (d), and (f) H8-L2; and (c), (e), and (g) ResUnet model retrieval in 2020. Color represents the occurrence frequency obtained by kernel density estimation.

it can be seen that the majority of CTH from the ResUnet model is close to the target (RMSE = 0.95 km and MAE = 0.91 km). As can be seen from Fig. 5(b), the majority of CER is mainly within the range of 0–60 μm (96.33% of all clouds), and the ResUnet model has a reliable retrieval result with an RMSE of 7.14 μm and an MAE of 6.38 μm. This is because the CER target values from H8-L2 are derived from the radiance of AHI near-infrared band 6 (central wavelength

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TABLE V
EVALUATION OF CLOUD PROPERTIES PRODUCTS FOR ALL, WATER, AND ICE CLOUDS USING MYD06 DATA IN 2020 AS THE BENCHMARK

CLP	Evaluation indicator	CTH (km)			CER (μm)		COT	
		FY4A-L2	H8-L2	ResUnet	H8-L2	ResUnet	H8-L2	ResUnet
All	RMSE	3.58	2.72	2.79	10.14	11.30	14.62	16.12
	MAE	2.16	1.74	1.81	6.58	7.98	6.64	8.23
	MBE	-1.17	-0.81	-0.66	0.61	0.64	0.76	0.87
Water	RMSE	1.74	1.53	1.60	4.82	6.20	10.77	13.33
	MAE	0.94	1..06	1.27	3.14	4.67	6.06	8.44
	MBE	-0.25	0.66	1.01	0.32	0.45	1.27	1.31
Ice	RMSE	3.27	2.38	2.11	10.84	11.04	13.32	14.89
	MAE	2.33	1.72	1.55	7.95	8.53	5.91	7.55
	MBE	-2.04	-1.23	-1.13	0.48	-0.20	1.74	1.89

at 2.4 μm), while the values retrieved by the ResUnet model use the BT of the FY4A/AGRI water vapor and longwave infrared bands (central wavelengths at 5.8–13.8 μm). The CER larger than 60 μm has little influence on the radiance values of water vapor and longwave infrared bands, resulting in greater retrieval difficulty [32]. In addition, we also note that the region near the diagonal range of 5–30 μm has the highest frequency. For samples with a target CER within 5–30 μm, the CER retrieval is more accurate with an RMSE of 4.19 μm.

As for the results for the retrieved COT given in Fig. 5(c), the ResUnet model estimates are highly consistent with COT target values in the range of 0–60 (RMSE = 9.01 and MAE = 6.83). However, the ResUnet model has an underestimation (MBE = −1.76) in the range of 0–150, and the RMSE and MAE are 19.12 and 11.54, respectively. The reason for this large error is that COT is sensitive to the radiance of the visible channel of the satellite sensor, and the target values from H8-L2 are retrieved from observations of the AHI visible bands 1–3 (central wavelengths at 0.46, 0.51, and 0.64 μm), but the FY4A/AGRI water vapor and longwave infrared bands BT (central wavelengths within 5.8–13.8 μm) are used in the ResUnet model. Moreover, we find that the ResUnet model performs well when COT is less than 20, giving an RMSE and MAE of 4.70 and 3.50, respectively. This analysis of the cloud properties retrieval on the testing dataset shows that the CTH, CER, and COT can be retrieved based on the ResUnet model using the BTs of FY4A/AGRI water vapor and longwave infrared bands.

Fig. 6 gives the joint probability density distribution of the CTH, CER, and COT products from FY4A-L2, H8-L2, and ResUnet model retrieval using MODIS cloud properties products as benchmarks. The valid ranges for official CTH, CER, and COT products from MYD06 are 0–17.5 km, 4–60 μm, and 0–150, respectively. As can be seen from Fig. 6(a)–(c), the CTHs of FY4A-L2, H8-L2, and ResUnet model retrieval cover 0–17.5 km and are close to the diagonal line although H8-L2 and the ResUnet model have a slight overestimation, and FY4A-L2 has a slight underestimation in the range of 0–3 km. Compared with MODIS, the CTH of FY4A-L2 has more underestimated pixels in the high value portion, resulting in a maximum MBE (−1.17 km) in three CTH products. The RMSEs (MAEs) of the three CTH products are 3.58, 2.72, and 2.79 km (2.16, 1.74, and 1.81 km), respectively, indicating that the results of ResUnet model retrieval and H8-L2 are closer to MODIS CTH than FY4A-L2 (Table V). For the CER products, both H8-L2 and ResUnet model have a higher frequency around the diagonal line within 40 μm [Fig. 6(d) and (e)]. The MBE (0.61 and 0.64 μm), RMSE (10.14 and 11.30 μm), and MAE (6.58 and 7.98 μm) produced by the two CER products are relatively close (Table V), with the errors mainly attributable to pixels larger than 40 μm. For the COT products, we find that H8-L2 and ResUnet model

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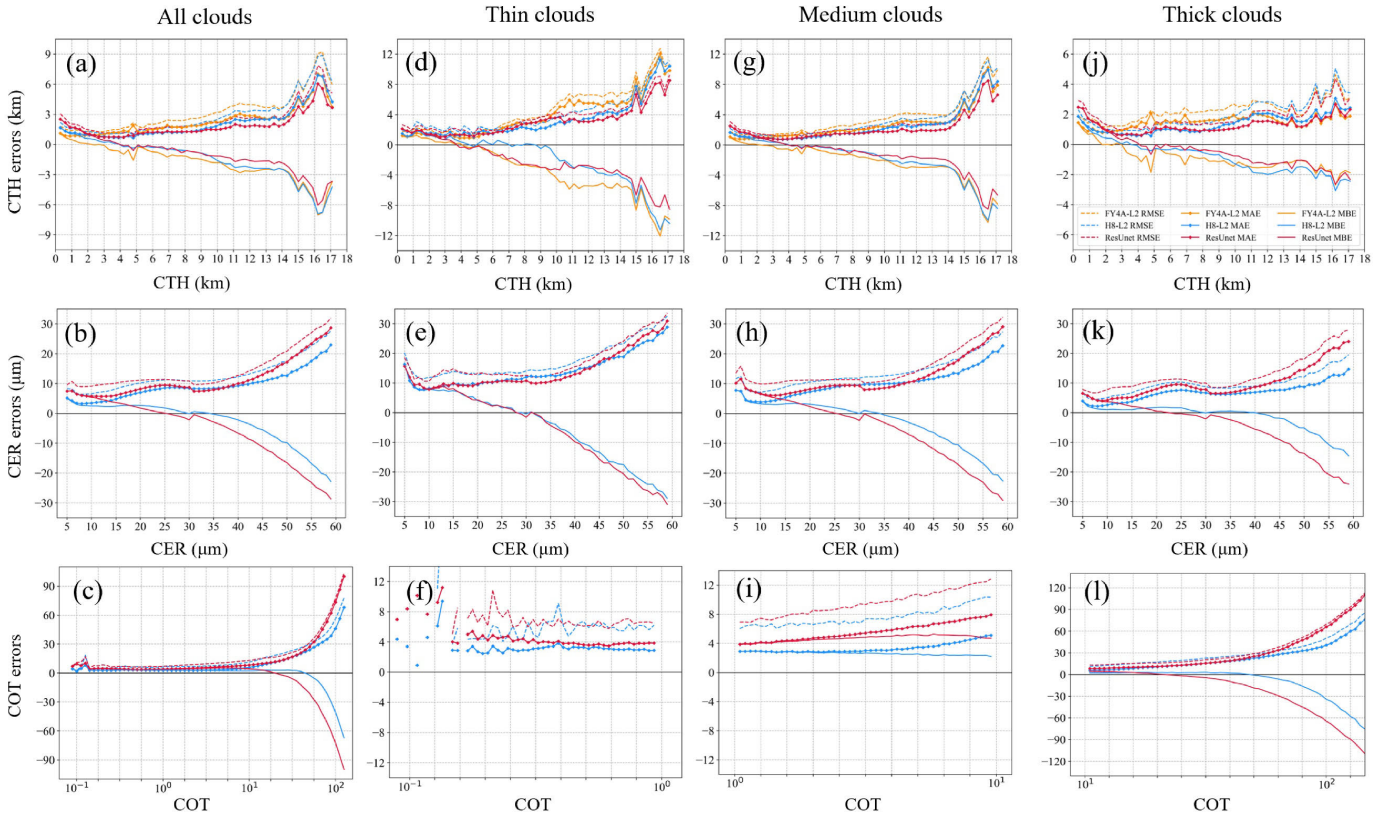


Fig. 7. Error curves of (a), (d), (g), and (j) CTH; (b), (e), (h), and (k) CER; and (c), (f), (i), and (l) COT products for (a)–(c) all; (d)–(f) thin; (g)–(i) medium; and (j)–(l) thick clouds from FY4A-L2 (orange lines), H8-L2 (blue lines), and ResUnet model retrieval (red lines) compared with official MODIS cloud products in 2020. Thin, medium, and thick clouds are divided based on COT values with thresholds of 1 and 10.

are most densely distributed around 10 on the diagonal line [Fig. 6(f) and (g)], and the two COT products exhibit a slight overestimation (MBE: 0.76 and 0.87) and similar accuracy (RMSE: 14.62 and 16.12; MAE: 6.64 and 8.23) in the range of 0–150 (Table V). In addition, in different seasons, RMSE of cloud properties products retrieved by the ResUnet model is close to H8-L2, and the error of CTH is significantly smaller than that of FY4A-L2. This indicates that the ResUnet model has a good generalization ability and a good stability in the seasonal cycle (Table SI). It is worth noting that, unlike ResUnet model retrieval and H8-L2, the CER and COT products of MODIS are retrieved from radiation in visible and near-infrared bands 1, 2, and 7 (central wavelengths at 0.65, 0.86, and 2.1 μm) and has different assumptions for different ice crystal scattering shapes in the algorithms. Due to differences in the satellite sensors' sensitivity to cloud properties for different radiation bands, the cloud properties of different products are also different. To better illustrate the applicability of cloud products, further evaluation is necessary.

We begin by discussing the quality of cloud properties products for different CLP. The evaluated results of water clouds, ice clouds, and all clouds are given in Table V. It can be seen that the RMSE (MAE) of the three CTH products is higher than 1.53 km (close to 1 km) for water clouds, and the ResUnet model gives the greatest overestimation of MBE (1.01 km). However, ice clouds of the three CTH products are clearly underestimated (MBE < -1 km), and the errors produced by the ResUnet model and H8-L2 are lower than those produced by FY4A-L2. The CER products of ResUnet model retrieval and H8-L2 have comparable precision for water and ice clouds, but the COT product of ResUnet model retrieval is slightly less precise than H8-L2. In general, the errors for ice clouds are larger than those for water clouds in the CTH and CER products, the reason for which is inseparable from the physical properties of clouds. Water clouds are composed primarily of water droplets with a small radius of about 10 μm, which are distributed mainly in the lower atmosphere where the temperature is higher. In contrast, ice clouds are composed primarily of ice crystals distributed in the upper atmosphere where the temperature is lower. Their optical thickness is generally thinner, which is why they have a wider error range.

A comparison analysis of the errors for thin, medium, and thick clouds in cloud properties products is shown in Fig. 7. From the error curves of the CTH products, we find that the ResUnet model has the lowest precision, being within 3 km. In the range of 3–17.5 km, the error curves of the ResUnet model and H8-L2 almost overlap and are distributed lower than FY4A-L2 when CTH $<$ 10 km, while the H8-L2 and FY4A-L2 products have larger errors when CTH $>$ 10 km, indicating that the ResUnet model has a higher precision than FY4A-L2 in this range [Fig. 7(a)]. There are also significantly larger errors (MAE $\approx$ 5 km) for thin clouds between 10 and 14.5 km in FY4A-L2 [Fig. 7(d)]. Additionally, the error distributions of medium and thick clouds for the three CTH products are relatively close. The errors for CTH products

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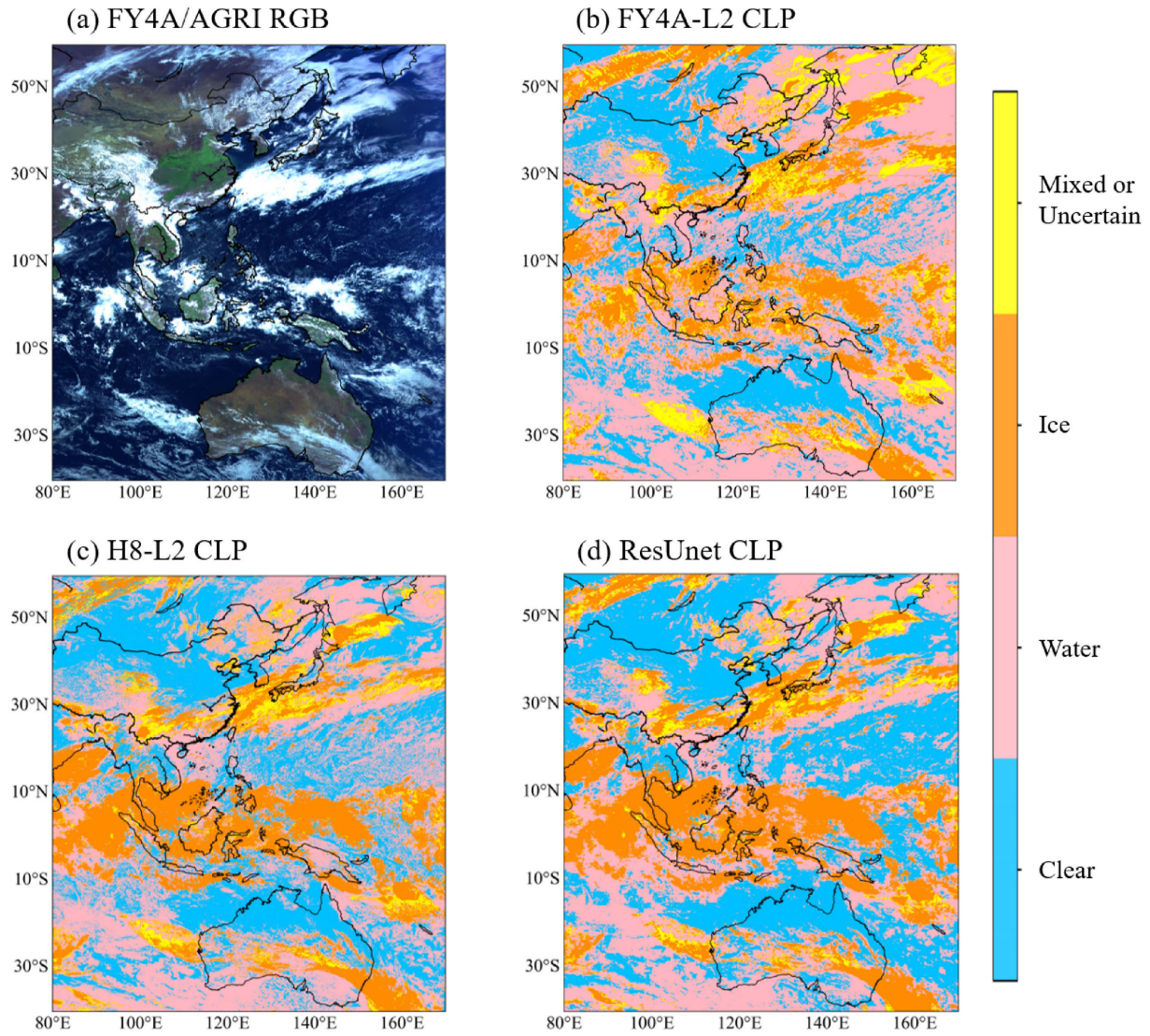


Fig. 8. (a) Natural color image composited of reflectance from FY4A/AGRI visible bands 1–3. Spatial distribution of CLP products from (b) FY4A-L2; (c) H8-L2; and (d) ResUnet model retrieval at 04:00 UTC on April 26, 2020.

mainly come from thin and medium clouds, especially for high clouds above 14.5 km. This is because observations of thinner clouds are more easily disturbed by surface information when retrieving CTH.

The errors for ResUnet model retrieval are significantly larger than those for H8-L2 when CER $>$ 40 μm, indicating that larger particles have a higher degree of underestimation [Fig. 7(b)]. For the retrieval of thin clouds, the ResUnet model performs as well as H8-L2, the average MAE and RMSE reach about 10 μm. For medium and thick clouds, only a CER smaller than 40 μm can achieve similar precision [Fig. 7(h) and (k)], so the errors for CER products mainly come from pixels larger than 40 μm. From Fig. 7(c), we can see that MBE of ResUnet model retrieval and H8-L2 is around 0 when COT $<$ 20, and MAE and RMSE increase significantly when COT $>$ 60. Moreover, thick clouds have relatively large errors compared with thin and medium clouds [Fig. 7(f), (i), and (l)]. This is especially so when COT is greater than 60, where the error is close to 30 (RMSE: 27.48 and 29.90). Therefore, the errors for COT products are mainly from optically thick clouds, and both of them have reliable and comparable precision in the range of 0–60 (RMSE: 8.25 and 9.96) and higher accuracy within 20 (RMSE: 3.87 and 5.25), which indicates that the image-by-image deep learning model can improve the reliability of COT estimation for opaque clouds to 60.

Because parallax error can affect space matching for multisatellite applications, it is important to correct this error to ensure that different sensors capture the same object in the same location. In addition, the CTH from the stationary satellite (FY4A and H8), as the radiation height, should be evaluated more accurately by using the detection data of active sensors of the CALIPSO satellite. Therefore, we evaluated the influence of parallax errors in different spatial extents and seasonal scales using CTH products with a spatial resolution of 1 km provided by CALIPSO. According to the longitude, latitude, and CTH information acquired by the CALIPSO satellite, we eliminate parallax errors between geostationary and polar orbit satellites by the geometric relationship [52], [53]. After

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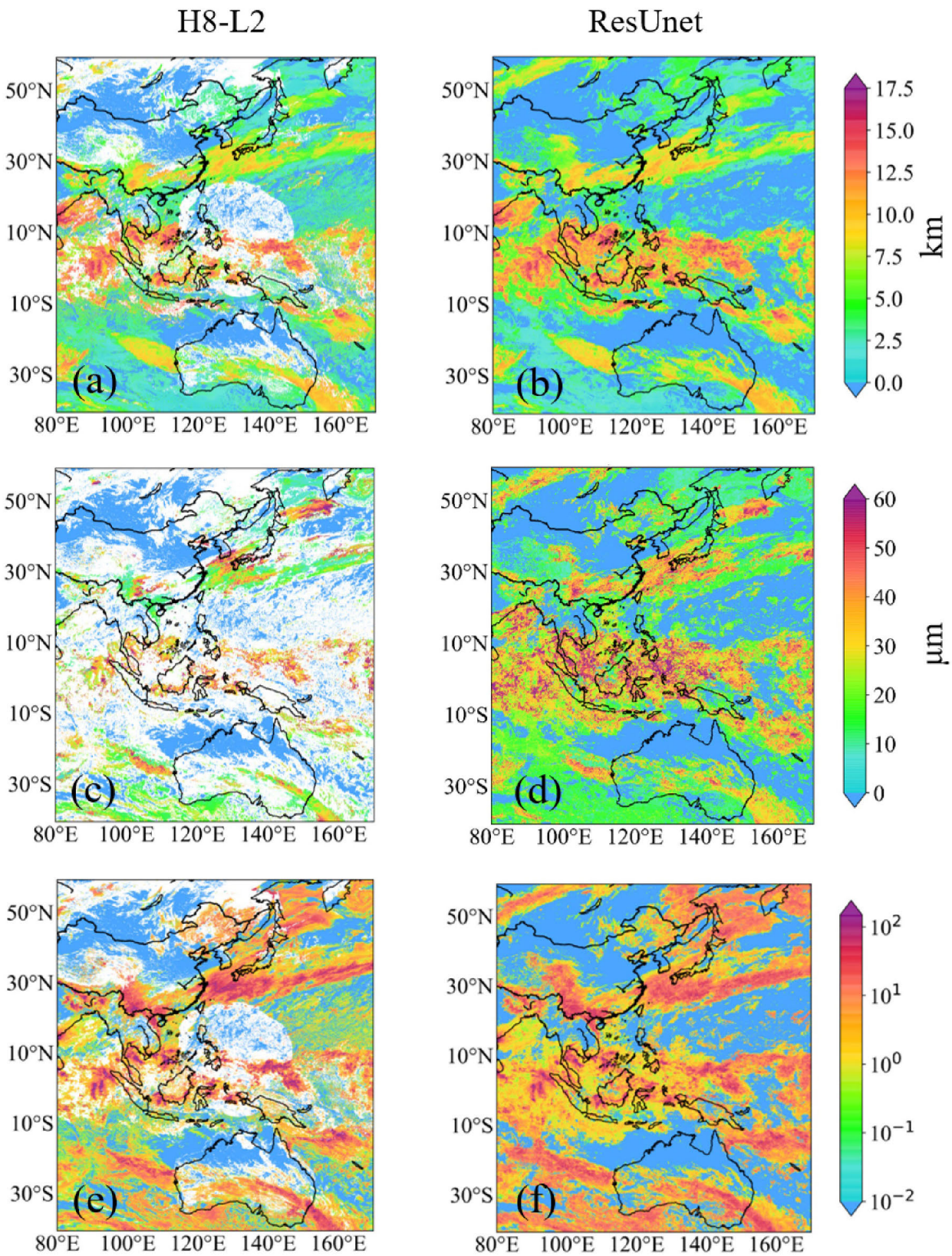


Fig. 9. Spatial distribution of (a) and (b) CTH; (c) and (d) CER; and (e) and (f) COT at 04:00 UTC, April 26, 2020. The left [(a), (c), and (e)] column is the target images from H8-L2, and the right [(b), (d), and (f)] column is the retrieval images of the ResUnet model.

parallax correction, we found that RMSE of CTH products from the ResUnet model, FY4A-L2, and H8-L2 decreased by 0.08, 0.06, and 0.09 km, respectively. In different seasonal scales, the RMSE of three CTH products decreased slightly (Table SIII). According to the RMSE distribution of CTH products in different SZA ranges, when SZA is smaller (<20°), the RMSE of CTH products is larger. This is because areas with smaller SZA are located near the

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equator and have stronger convection, resulting in generally higher CTH, while high latitudes with larger SZA have lower CTH due to weaker convection. Although the parallax error will increase with the increase of SZA and CTH, the influence of parallax error in different spatial ranges is not significant due to the smaller CTH in the region with larger SZA (Table SII). In addition, since the spatial resolution of the FY4A-L2 and H8-L2 cloud products is relatively coarse (4 and 5 km), and the SZA matching between MODIS data and CALIPSO satellite data is small, the parallax effect is not significant [30].

In general, we also found that the RMSE of CTH retrieval by the ResUnet model was all smaller than FY4A-L2 in different SZA ranges and the seasonal cycle. This also fully illustrates that cloud products retrieved by the ResUnet model can improve the CTH precision for the FY4A satellite, which is consistent with the assessment results obtained using MODIS data as a benchmark (Fig. 6 and Table V).

D. Case Analysis

The retrieval result from 04:00 UTC on April 26 is selected as a case for evaluating the performance of the ResUnet model across the region spanning 80°E–170°E and 40°S–60°N. First, we compare and analyze the cloud detection and identification results of CLP products obtained by FY4A-L2, H8-L2, and ResUnet model retrieval. Fig. 8(a) shows the true color (RGB) image taken by FY4A/AGRI. According to the spatial distribution of CLP in Fig. 8(b)–(d), we find that the ResUnet model performs well when capturing cloud contours compared with the target image from H8-L2. Moreover, H8-L2 and the ResUnet model have significantly better correspondence for the cloudy region in RGB images than FY4A-L2. This is particularly so for mid-latitudes near 80°E–170°E, where there are a large number of confused pixels between cloudy and clear sky in the products of FY4A-L2. However, the ResUnet model can distinguish them well. As for clear sky pixels being misjudged as water clouds, the distribution of water cloud pixels in FY4A-L2 is significantly greater than those in H8-L2 and ResUnet model retrieval. In the cloudy region near the equator (10°N–10°S), the CLP distribution of H8-L2 and ResUnet model retrieval is dominated by ice clouds (>53% of pixels), while FY4A-L2 has only 37.22% of pixels classified as ice clouds near the equator. In the whole region, the proportion of mixed or uncertain clouds in FY4A-L2 is significantly higher than that in H8-L2 and ResUnet model retrieval, which results in distinctly different contour characteristics for ice clouds and water clouds. Overall, the ResUnet model can significantly improve the accuracy of FY4A-L2 in cloud detection and identification.

To verify the ability of the ResUnet model to capture the characteristics of cloud properties, we also analyze the spatial distributions of CTH, CER, and COT at UTC 04:00 on April 26, 2020. As shown in Fig. 9, the left column shows target images from H8-L2 official cloud products with missing pixels (unfilled) due to the use of shortwave channel radiation, and the right column shows the retrieval images without missing pixels due to the use of thermal infrared BT in the ResUnet model [24], [25]. In addition, the distribution of cloud properties retrieved by the ResUnet model has a good correspondence with those from H8-L2, and it is complete and continuous, indicating that the ResUnet model can be used to complement the missing values of cloud properties products obtained by the H8-L2 official algorithm. If we compare the ResUnet model with the target image for CTH in Fig. 9(a), we find that it can capture the distribution of high and low clouds well [Fig. 9(b)]. In the spatial distribution of CER, we notice that most of the values of CER in the target images are distributed within the range of 0–60 μm [Fig. 9(c) and (d)]. Furthermore, there is an obvious overestimation in the region near the Northwest Pacific, which is mainly due to CLP misjudgment in Fig. 8(d). It should be noted that all clouds can be classified into thin, medium, and thick clouds based on COT values with thresholds of 1 and 10. Compared with the target image [Fig. 9(e)], the retrieval image in Fig. 9(f) shows that the ResUnet model has reasonable estimation and discrimination for medium and thick clouds, but some thin clouds will be identified as either clear sky or thick clouds in parts of the Northwest Pacific (120°E–180°E, 5°N–20°N) and Eastern Indian Ocean (80°E–110°E, 10°S–30°S), which is also affected by misjudgment of the CLP. In general, the ResUnet model exhibits a good performance in the retrieval of cloud properties.

V. Conclusion

We proposed a deep-learning algorithm that performs image-by-image and all-weather retrieval for the CLP, CTH, CER, and COT products of the FY4A satellite. The ResUnet model can achieve daytime and nighttime cloud retrieval over East Asia with the BT from FY4A/AGRI water vapor and longwave infrared channels as input features. Moreover, the ResUnet model takes H8-L2 cloud products as true targets for image-by-image learning, which can make up for the difficulty of retrieving thick clouds by thermal infrared radiation at night as much as possible. To demonstrate the rationality and interpretability of the input features, we investigated the relative contribution of each feature in the ResUnet model after data preprocessing. We found that BT plays the most important role in the identification of CLP, while meteorological fields and SZA greatly improve the performance in retrieving cloud properties. Furthermore, the addition of CLP enables better capture of the spatial characteristics of cloud properties.

To retrieve cloud properties more accurately, we first retrieved and identified the CLP of FY4A/AGRI using the ResUnet model. Using the testing dataset for 2020, OA can reach 90.62%, and for clear sky and ice clouds, it can achieve an accuracy of over 93%. Then, using the CLP product from the CALIPSO satellite as a benchmark, we found that the OA of FY4A-L2 is less than 82% for cloud detection, while the OA of H8-L2 and ResUnet model retrieval can reach 86.57% and 84.09%. Moreover, when identifying clear sky, water clouds, and ice clouds, the OA of ResUnet model retrieval and H8-L2 is 76.74% and 78.97%, respectively, which is an improvement of the 71.81% for FY4A-L2. Therefore, we can

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say with confidence that the new CLP product retrieved by the ResUnet model significantly improves cloud detection and CLP identification.

Next, we added the retrieved CLP product into the ResUnet model to test its ability to retrieve cloud properties. On the testing dataset, the retrieval of CTH has high accuracy with an RMSE (MAE) of less than 1 km. However, the different sensitivities of cloud properties to BT in different bands can lead to larger errors in the high values of CER and COT. In the range of 0–60, the RMSE (MAE) of the CER and COT retrieval is 7.14 μm and 9.01 (6.38 μm and 6.83), respectively, indicating that the ResUnet model performs reasonably well. Taking MYD06 cloud properties products from MODIS as benchmarks, we evaluated the quality of FY4A-L2, H8-L2, and ResUnet model retrieval from multiple perspectives. The joint probability density distributions and error distributions of thin, medium, and thick clouds in different value domains show that the ResUnet model has slightly lower precision than H8-L2 and higher precision than FY4A-L2 for CTH products, and that the errors mainly come from thin and medium clouds. The CER products from ResUnet model retrieval and H8-L2 have similar precision within 40 μm and better quality for medium and thick clouds. For COT products, the errors are mainly produced in thick clouds. The precision of ResUnet model retrieval and H8-L2 is reliable and comparable in the COT range of 0–60, and is higher within 20. The results show that the reliability of COT estimation for opaque clouds can be improved to 60 by using the ResUnet model with image-by-image learning way.

The deep-learning model has a higher accuracy and efficiency compared with traditional physical retrieval algorithms. Experiments have shown that it takes about 1 min to conduct a single retrieval across the whole study area, which can be sufficient to retrieve high-precision and all-weather cloud products retrieval for FY4A/AGRI. In addition, different network structures and parameter settings have different effects on the model, so we experimented with changing parameters and using samples with different distributions during training to determine which combination makes for the best model (Table SIV). For example, we selected 200 000 random samples and 91 181 samples with four phase types for training the CLP model, but the OA on the same testing dataset with 50 000 random samples was lower than that of the 159 223 samples used in this study. We also used a OneCycleLR learning rate, changed the batch size and epoch, and used label distribution smoothing to overcome the difficulties from having an unbalanced data distribution, but none of these realized ideal the model performance. The accuracy of the CER model in particular drops significantly when the sample size of the training dataset is increased. As for the retrieval of CER and COT, we also used the spatiotemporal sequence prediction model-SmaAtUnet, which adds an attention mechanism based on Unet and uses depthwise-separable convolution to replace traditional convolution. There is, however, not much of an improvement in accuracy compared with the ResUnet model. Of course, there may be better choices for the model structure and parameter settings, but further work is needed to determine what they might be.

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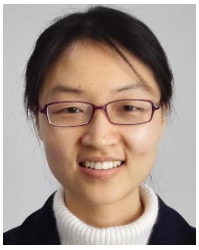
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